

Unit

3

Lesson

2



Beyond Imagination.

Boost Your App's Senses.

Time Frame: [90 min].

Ready, Set, Learn!

You Need to Know

- Improve the smart glasses prototype by integrating extra sensors to increase safety and usefulness.
- Apply logical reasoning to detect and correct mistakes in measurement, scaling, or how the app responds.

Challenge questions of the lesson.

1. How can we improve our smart glasses to be even safer and more helpful for users?

SEL Relation

1. Social Awareness (Empathy):

- Students think about how to make smart glasses safer and more helpful for others, especially people who may have difficulty seeing or moving safely.

2. Responsible Decision-Making:

- Students choose the right sensors and test their solutions carefully, making sure they work correctly and solve real problems.

3. Self-Management (Perseverance):

- When something doesn't work, students try again, fix mistakes, and keep improving their design without giving up.

4. Collaboration:

- If working in pairs or groups, students share ideas and help each other solve technical problems thoughtfully.

Engage



Teacher Role:

- **Story Connector:** Begin by reading Scene 1 and Scene 2 aloud with expression. Pause to discuss the problems the students in the story noticed.
- **Facilitator of Critical Thinking:** Guide students to reflect on what went wrong in the smart glasses test and why improving sensor placement matters.
- **Inquiry Coach:** Encourage students to brainstorm sensor placement and think about how the app might need to change.
- **Support Collaborator:** Observe group discussions and prompt deeper thinking by asking:
 - “What happens if a person turns their head?”
 - “How will you make sure the glasses ‘see’ enough?”

Expected From Students:

- **Reflection and Analysis:** Think about the current limitations in the smart glasses' design based on the story.
- **Creative Problem-Solving:** Work in pairs or teams to sketch new placements for additional sensors (e.g., sides or top of the frame).
- **Logical Planning:** Begin thinking about what changes in the code or app would be needed for the new sensors to work properly.
- **Communication:** Share their ideas with the class and explain how their design helps improve safety and usability.

Parts of the books included.

After printing the frame, they tested the glasses indoor.

Adam walked forward.
“The sensor only detects what’s in front, not the sides.”

Laila bumped into a box. “I walked into something the sensor didn’t catch.”

Jamila pointed to the side.
“We need more sensors for better coverage.”



Ms. Sara: “Now, kids, you’ll need to make some modifications to fit your new changes.”

Adam: “We have to make enough room in the glasses for all these sensors.”

Laila: “And we’ll need to update our code to match the new setup!”

Pyramakerz

Let's Think!



Time for a new challenge!

The smart glasses work—but something’s not quite right. When Adam walked forward, everything was fine... but Laila still bumped into something on the side!

Uh-oh! It seems like the glasses can’t “see” everything around them.

What if we could give the glasses side vision, like super senses?

Your task:

Work with your team to improve the glasses by adding extra sensors. Think carefully—where should you place them to make the glasses safer and smarter?

Story



1. Set the Scene:

- Begin by reminding students of their previous smart glasses design and testing process. Then say:
 - “Today, we’ll explore how to make our smart glasses even smarter! But first, let’s listen to what happened when the students in our story tested their design.”

2. Read the Story Aloud:

- Read Scene 1 and Scene 2 aloud with expression, acting out the parts if appropriate. Pause after each student’s line to ask:
 - “What problem did they notice?”
 - “Why didn’t the sensor catch everything?”

3. Prompt Curiosity and Empathy:

- Ask:
 - “How would you feel if your smart glasses missed something important and let you bump into something? What could you do to fix that?”

4. Introduce the Challenge:

- Explain that their mission today is to improve the smart glasses by:
 - Adding more sensors.
 - Adjusting their placement.
 - Updating their code logic.
- Say:
 - “Just like Adam, Laila, and Jamila, you are the engineers. Let’s figure out how to make smart glasses that are truly smart—from all sides!”

Possible Strategies for Implementation:

1. Dramatic Reading (Role Play):

- Assign students to read the dialogue parts of Adam, Laila, Jamila, and Mrs. Sara aloud with expression. This builds engagement and helps them empathize with the characters’ design challenge.

2. Storyboard Visualization:

- After hearing the story, ask students to draw a quick 3-panel comic showing what went wrong with the glasses and how they might improve them.

3. Think-Pair-Share:

- Think: Individually, students reflect on what went wrong in the scene.
- Pair: Discuss with a partner what could be changed.
- Share: Pairs share their ideas with the class to build collective understanding.

4. Problem Framing Sentence Starters:

- Guide students to reframe the challenge using prompts like:
 - “The problem was...”
 - “We need to improve the glasses by...”
 - “Sensors should be placed...”

5. "Zoom-In Inquiry":

- Pause after each character’s line and ask:
 - What does this tell us about the current design?
 - What new need is being revealed?
 - This builds curiosity and problem-identification skills.

6. Class Anchor Chart:

- Create a large “Problem & Solution Wall.”
 - Under “Problems Found in the Story,” write what went wrong.
 - Leave space under “Possible Fixes” for students to add sticky notes with ideas.

Let's Think!



Objective:

Students will analyse the limitations of their initial smart glasses design and collaboratively propose sensor placement improvements to enhance safety and functionality.

Answers:

1. Add side sensors to detect obstacles on the left and right.
2. Place sensors near the edges of the glasses’ frame.
3. Add a buzzer or LED for side alerts.
4. Tilt some sensors slightly for wider field coverage.

Possible Strategies for Implementation:

- **Prompt with Scenario Recap:** Re-read the story or act it out to ensure students understand the challenge.
- **Group Brainstorming:** In small groups, sketch new sensor placements and justify their ideas.
- **Use Real-World Analogies:** Compare glasses to animal senses (like owls or cats with side vision).
- **Design Thinking:** Have students go through empathize → define → ideate steps using this problem.
- **Sticky Note Mapping:** Provide blank glasses templates where students can place sticky notes representing sensors.

- **Think-Pair-Share:** Students reflect individually, then discuss in pairs, and finally share aloud their improved design idea.

Explorer's Toolkit



Teacher Role:

- **Facilitator of Design Thinking:** Guide students through reflective questions on sensor placement and logic.
- **Prompting Engineer Mindsets:** Encourage students to think like problem-solvers—testing, analyzing, and iterating their design.
- **Support With Tools:** Assist with navigating the Tinkercad environment and interpreting the wiring diagram.
- **Encourage Peer Review:** Promote collaboration by encouraging teams to test and share designs with each other.

Expected From Students:

- Use critical thinking to determine the most effective placement of side sensors.
- Test sensor spacing and configurations using Tinkercad circuit simulation.
- Discuss and document what happens when the sensors are poorly positioned.
- Explain how the app will respond to sensor triggers (e.g., alert, visual cue, sound).
- Apply engineering reasoning to justify their design decisions.

Parts of the books included.



DIY Lab

Now that you've decided to add two more ultrasonic sensors to your smart glasses, it's time to think like real engineers!

Ask yourself:

- Where exactly should the sensors go so they can detect obstacles from the sides?

- What happens if the sensors are too close together—or too far apart?

- How will the glasses know which sensor is detecting something?

- What should the app do when a side sensor is triggered?

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Try It Out!

Test different sensor positions in your Tinkercad design. Use your reasoning to:

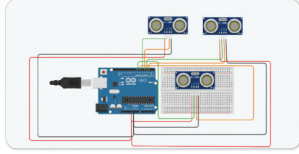


Figure 27: Multi-Sensor Ultrasonic Circuit Connection.

- Upload the code and Compare which sensor setup gives the best coverage.

Fill in the blank to complete the code.

```
// Front sensor pins
const int trigFront = 2;
const int echoFront = 3;

// Left sensor pins
const int trigLeft = 4;
const int echoLeft = 5;

// Right sensor pins
const int trigRight = 6;
const int echoRight = 7;

// Pin definitions
const int distanceFront, distanceLeft, distanceRight;

void setup() {
  // Set all pins as OUTPUT
  pinMode(trigFront, OUTPUT);
  pinMode(echoFront, INPUT);
  pinMode(trigLeft, OUTPUT);
  pinMode(echoLeft, INPUT);
  pinMode(trigRight, OUTPUT);
  pinMode(echoRight, INPUT);
}

// Add some timing
Serial.println("-----");
delay(1000);

// Function to measure distance from a sensor
int getDistance(int trigPin, int echoPin) {
  digitalWrite(trigPin, LOW);
  delayMicroseconds(2);
  digitalWrite(trigPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(trigPin, LOW);
  duration = pulseIn(echoPin, HIGH);
  int distance = duration * 0.034 / 2;
  return distance;
}
```

Figure 28: Arduino Code for Multi-Sensor Distance Detection.

STEAM SCOUTS

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- Adjust your circuit and code to respond to each sensor correctly.

- Solve problems if the sensors give confusing or overlapping signals.

Objective:

Students will test, modify, and optimize their smart glasses by adding additional ultrasonic sensors, refining the placement, circuit wiring, and code to improve obstacle detection and avoid signal interference.

Materials Needed:

- Computer with Tinkercad access.
- Arduino Uno, breadboard (virtual or real).
- Ultrasonic sensors (3 or more).

Answers:

1. Where exactly should the sensors go so they can detect obstacles from the sides?

- The sensors should be placed on the left and right arms of the glasses, angled slightly outward so they can detect obstacles to the sides of the user.

2. What happens if the sensors are too close together—or too far apart?

- If the sensors are too close, they might interfere with each other's signals and give incorrect readings.
- If they're too far apart, they may miss objects between them or cause blind spots in detection.

3. How will the glasses know which sensor is detecting something?

- Each sensor is connected to a different pin on the Arduino. The code reads them separately and uses variables to identify which sensor is triggered.

4. What should the app do when a side sensor is triggered?

- The app should alert the user with a clear message like "Obstacle on your left" or "Watch out on the right" using sound or text-to-speech output.

Code Completion Answers:

- Front sensor pins

```
const int trigFront = 9;  
const int echoFront = 8;
```

- Left sensor pins

```
const int trigLeft = 5;  
const int echoLeft = 4;
```

- Right sensor pins

```
const int trigRight = 3;  
const int echoRight = 2;
```

- Obstacle detection print line:

```
Serial.println("Obstacle on the right");
```

Activity Instructions:

1. Observe and Reflect: Read the story of students testing their glasses and discovering flaws in sensor placement.
2. Plan Your Solution:
 - Answer guiding questions about placement, distance between sensors, detection logic, and app response.
3. Prototype & Simulate:
 - Add side ultrasonic sensors in Tinkercad and wire them.
 - Use the guide image for basic circuit layout.
4. Code & Debug:
 - Use the provided Arduino sketch. Fill in the missing code parts.
 - Test how the glasses behave when each sensor is triggered.
5. Adjust and Optimize:
 - Calibrate sensor distances in code.
 - Handle overlapping signals or faulty triggers.
6. Compare Results:
 - Upload code variations and test different configurations.
 - Decide which sensor setup gives the best coverage.

Possible Strategies for Implementation:

- **Think-Aloud Modeling:** Demonstrate adding one sensor and coding its logic as a group before students try.
- **Peer Debugging:** Let students exchange circuits or code with a partner to review for errors.
- **Design Rationale Chart:** Have students complete a chart justifying where and why they placed each sensor.
- **Code Puzzle:** Mix up the code blocks and let students reassemble them logically.
- **Reflection Writing:** Ask students to explain how their changes improved the design and what challenges they overcame.

Explain



Teacher Role:

- Introduce the idea of logical reasoning and connect it to the students' recent work with sensor placement and debugging.
- Guide students to watch the video attentively and think about how the examples relate to problem-solving in tech projects.
- Lead a class discussion afterward to reflect on what logical reasoning looks like in real life and in their smart glasses design process.

Expected From Students:

- Watch the video and understand the concept of logical reasoning.
- Identify how they used logical thinking in sensor placement and code correction.
- Share examples of logical reasoning from the video or their own experience.

Parts of the books included.



Screen Lessons

In our last activity, we used something called logical reasoning!
Click the link to explore more about how logical reasoning helps us make smart decisions!

<https://study.com/academy/lesson/video/logical-thinking-reasoning-questions-lesson-for-kids.html>



Screen Lessons

Objectives:

- Understand what logical reasoning is and how it supports problem-solving.
- Apply logical reasoning to improve designs and fix sensor-related coding issues.
- Reflect on decision-making steps in engineering tasks.

Video Link:

<https://study.com/academy/lesson/video/logical-thinking-reasoning-questions-lesson-for-kids.html>



Possible Strategies for Implementation:

1. Pre-Watching:

- Ask students: “What do you think logical reasoning means?”
- Share a quick scenario (e.g., sensors giving wrong readings) and ask how they’d figure out the problem.

2. During Watching:

- Ask students to write down 1–2 examples of logical reasoning shown in the video.
- Pause briefly to discuss key moments.

3. Post-Watching:

- Reflect together: “How did we use logical reasoning in our smart glasses project?”
- Optionally, do a Think-Pair-Share: students describe a time they used logical thinking in or outside class.

Elaborate



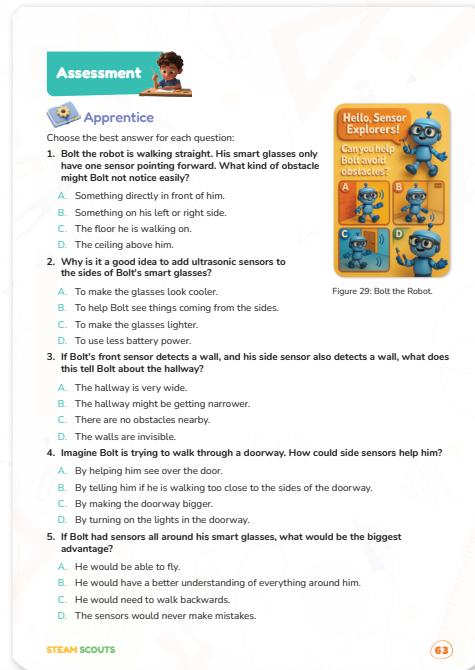
Teacher Role:

- **Assessment Facilitator:**
 - Guide students through the multiple-choice assessment. Remind them to think carefully and apply what they’ve learned about ultrasonic sensors, obstacle detection, and logical reasoning.
- **Observer & Supporter:**
 - Circulate as students work to offer clarification without giving answers. Encourage students to justify their choices if they’re unsure.
- **Feedback Provider:**
 - Review answers with the class and discuss reasoning, especially if there’s confusion. Highlight how these questions relate to real-world sensor design.

Expected From Students:

- Read and understand each question carefully.
- Apply knowledge from the smart glasses project, especially about sensor placement and reasoning.
- Choose the best answer and be ready to explain why it makes sense based on their understanding.

Parts of the books included.



Apprentice

Objective:

Assess students' understanding of how ultrasonic sensors function and how logical reasoning helps improve sensor placement and safety features in smart glasses.

Answers:

1. b) Something on his left or right side.
 - The glasses only have a front-facing sensor, so side obstacles won't be detected.
2. b) To help Bolt see things coming from the sides.
 - Side sensors increase coverage and prevent side collisions.
3. b) The hallway might be getting narrower.
 - If both front and side sensors detect walls, the space may be tight or closing in.
4. b) By telling him if he is walking too close to the sides of the doorway.
 - Side sensors help maintain safe spacing when passing through tight areas.
5. c) He would need to walk backwards.
 - The question is meant to be silly—full coverage doesn't help you walk backwards unless you're facing that direction.

Activity Instructions:

1. Students read each question carefully.
2. Choose the best answer based on what they learned during the smart glasses lessons.

3. Optional: Ask students to explain their reasoning for each answer orally or in writing.
4. Teacher collects and reviews answers to check comprehension and identify areas needing reinforcement.

Possible Strategies for Implementation:

1. Pre-Assessment Review:

- Review key vocabulary (sensor, obstacle, coverage, etc.).
- Briefly recap the sensor challenges the students worked on earlier.

2. During Assessment:

- Allow independent thinking first.
- Then use Think-Pair-Share to encourage discussion on tricky questions.

3. Post-Assessment:

- Go over answers as a class and discuss the logic behind each.
- Use misconceptions to launch a mini-review or quick experiment.

Evaluate



Teacher Role:

- **Guide Assembly:** Help students carefully transfer their Tinkercad circuit design into a real-world setup using sensors, wires, and a breadboard.
- **Ensure Safety:** Supervise use of electronics and tools, making sure students handle components safely.
- **Encourage Testing & Iteration:** Support students in testing their build, spotting issues, and thinking critically to improve performance.

Expected From Students:

- **Build the Circuit:** Recreate the circuit they designed in Tinkercad using real components like ultrasonic sensors and a Bluetooth module.
- **Assemble the Prototype:** Attach the built circuit securely to their 3D-printed smart glasses frame.
- **Test & Reflect:** Use the app to test the functionality of their smart glasses, checking for sensor coverage and responsiveness, and adjusting if needed.

Parts of the books included.



Showcase

Now it's time to bring your design to life!

Take the circuit you made in Tinkercad and build it using real parts.

Connect the ultrasonic sensors and Bluetooth module, then carefully attach everything to your smart glasses frame and test using the app!

Objective:

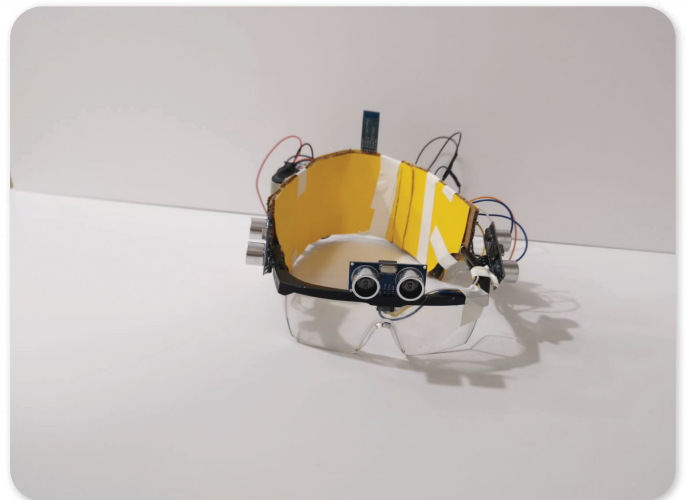
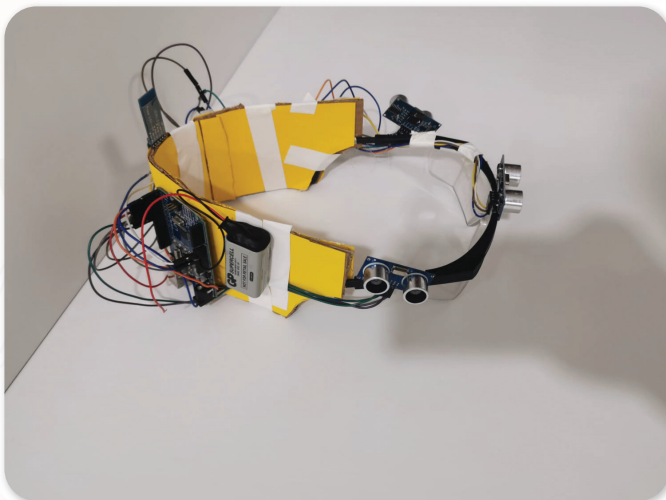
- To assemble and test a fully functional smart glasses prototype by connecting physical components and evaluating its real-world performance.
- Full project resources: https://docs.google.com/document/d/1xCnRHZX0X894v7aR_BwtnuEa0xVLMsa/edit



Activity Instructions:

1. Gather all needed components: ultrasonic sensors, Bluetooth module, Arduino, wires, and your 3D-printed glasses frame.
2. Rebuild the circuit from Tinkercad using real parts.
3. Connect the sensors and Bluetooth module securely to the glasses frame.
4. Upload the code and test your prototype using the mobile app.
5. Troubleshoot and improve the placement or wiring if needed.
6. Share and explain how your final model works.

Final Prototype:



Prototype video:

<https://drive.google.com/drive/folders/1Fkjhd7hpljaLLFjWWW0TM6gHssqdM7v>



Showcase Rubric:

Criteria	Excellent (3 Point)	Satisfactory (2 Points)	Needs Improvement (1 Point)
Circuit Build	All components connected correctly and securely.	Most components connected correctly.	Many errors in circuit or loose connections.
Sensor Functionality.	All sensors respond accurately.	Some sensors respond as expected.	Sensors do not respond properly.
Integration with Frame.	Frame holds components neatly and safely.	Some parts secure, others loose.	Poor or no integration.
Testing & Debugging.	Tested thoroughly; solved issues.	Basic testing done; minor issues.	Little or no testing.
Presentation & Explanation.	Clearly explains how it works.	Gives a basic explanation.	Struggles to explain prototype.

It's time to move to the improve part and add this.

Improve

We improved our smart glasses by adjusting the sensor positions after noticing that the left side wasn't detecting obstacles well. We repositioned the sensor outward and checked all the wiring to make sure it was secure. Then, we updated the code to match the new setup. After these changes, the glasses could detect obstacles on all sides more accurately and send alerts properly to the app.

Final Scene (Story)

- Ending the Theme – "A Walk with Confidence"
- Set a Reflective and Celebratory Tone.
- The teacher gathers the class and reads the final story scene aloud with emotion and pride. After finishing, the teacher pauses and says:
 - "You've just heard how your ideas and hard work made a real difference for someone. You didn't just learn about sensors or apps—you created something that gave someone freedom. That's the power of technology used with empathy."

Possible Strategies for Implementation.

1. Reflective Circle (Whole-Class or Small Groups).

- Ask students to sit in a circle and answer prompts such as:
 - "What part of this project made you most proud?"
 - "What was one thing you learned—not just about tech, but about helping others?"
 - "How did it feel to see your work make someone's life easier?"

2. Vision Wizard Certificates.

- Hand out personalized certificates (e.g., "Certified Vision Wizard – For Empowering Others Through Technology") as a symbolic celebration.

3. Gallery Walk

- Set up a mini exhibition where students display their app designs, circuits, and improvement journals. Parents, other classes, or school leaders can be invited to view and give feedback.

4. "Letter to Mr. Evans" Writing Activity.

- Students write a letter to Mr. Evans describing how they felt during the project and what they hope he enjoys most about the smart glasses.



Now I Can

Objective:

Students will improve their smart glasses prototype by integrating additional sensors for enhanced safety and functionality. They will also apply logical reasoning to identify and correct technical issues in measurement, scaling, or app behavior.

Activity Instructions:

1. Review the Prototype:
 - Begin by testing the initial smart glasses design with only the front sensor.
 - Observe any safety gaps—especially on the sides.
2. Problem-Solve Through Dialogue & Code:
 - Use student reflections and guided questions to identify where extra sensors should go.

- Modify both the 3D design (in Tinkercad) and Arduino code to accommodate new sensor inputs.
3. Test and Troubleshoot:
- Upload updated code and test in Tinkercad circuits or with real components.
 - Use logical reasoning to find if the sensors are too close or too far, or if outputs are incorrect.
 - Debug and improve the system by rechecking code structure, sensor alignment, and app behavior.
4. Document Improvements:
- Record all changes, reasons for placement choices, and any logical steps taken to resolve coding or design errors.
5. Reflect and Share:
- Use the “Now I Can” box as a personal or group reflection.
 - Share outcomes with the class or a partner using a demonstration or explanation of improvements made.

Relation to Standards:

1. ISTE Standards for Students:

- **1.4 Innovative Designer:**
 - Students use a design process to generate and test solutions, refining their smart glasses to make them safer.
- **1.5 Computational Thinker:**
 - Students apply logical reasoning and debugging skills to correct coding or sensor placement issues.

2. CSTA (Computer Science Teachers Association):

- **2-AP-13:**
 - Decompose problems and modify existing code to add functionality (e.g., improving sensor logic).
- **2-AP-17:**
 - Systematically test and debug programs using logical reasoning.
 - Cambridge ICT Starters / Digital Literacy:
- **Developing Ideas and Making Things Happen:**
 - Use ICT tools (like Tinkercad and Arduino) to create and improve digital systems.
- **Finding and Correcting Errors:**
 - Recognize and fix mistakes in digital projects.
- **UNESCO ICT-CFT:**
- **Technology Literacy Approach:**

- Apply digital tools to solve real-world problems (assistive technology for vision support).

3. Common Core (Math/Science):

- **CCSS.MATH.PRACTICE.MP1** – Make sense of problems and persevere in solving them.
- **NGSS Engineering Design (MS-ETS1):**
 - Define criteria and constraints of a design problem and evaluate solutions.



Mission Innovate

Final Mission – Design for Vision and Safety!

Overview:

This is a culminating, open-ended project that allows students to apply all the skills they've learned throughout the unit. Students will take ownership by identifying a real-world problem faced by people who have difficulty seeing and designing a smart tech solution that improves daily life and safety.

Project duration: 2 weeks

Teacher's Role:

- **Facilitator of Empathy & Inquiry:**
 - Begin by reading the mission prompt aloud. Encourage students to reflect on the needs of people who are blind or visually impaired.
 - Guide a discussion on what makes it hard to navigate public or private spaces without sight.
- **Prompter of Creative Solutions:**
 - Support students in identifying meaningful problems, such as crossing streets, locating doors, or recognizing obstacles.
 - Encourage brainstorming and ideation using sensors, sounds, vibrations, or visual/audio alerts.
- **Supporter of Student-Led Innovation:**
- Let students lead their own project journey. Provide tools, materials, and feedback when needed.
 - Remind them to draw on what they learned about coding, sensors, modeling, and app feedback.
- **Reflection Guide:**
 - Use the critical thinking questions and design rubric to help students evaluate their solutions.
 - Encourage them to iterate based on testing and peer feedback.
- **Assessment Guide:**
- Use the rubric to assess each team's project based on how well it identifies a clear

need, the creativity and functionality of the solution, technical accuracy (e.g., code or sensor use), and how effectively the prototype communicates information.

Implementation Tips:

- Let students work in pairs or small design teams.
- Ensure access to devices/tools (e.g., PictoBlox, Arduino, sensors, buzzers, LEDs, craft supplies).
- Encourage design journaling to capture steps, ideas, and iterations.
- Include time for showcasing designs to classmates or a guest visitor (e.g., a school staff member who tests the designs).
- Emphasize empathy, creativity, and problem-solving over perfection.